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Experimental investigation and thermodynamic assessment of Nd–H and Nd–Ni–H systems

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ABSTRACT

A coupled experimental investigation and thermodynamic study of the Nd–Ni–H system have been carried out and the self-consistent thermodynamic database on this system has been obtained. High pressure differential scanning calorimetry (HP-DSC) is used to determine the temperature of reaction $NdNi_5H_3 \leftrightarrow NdNi_5 + 1.5H_2$ in the pressure range of 3.01–11.01 MPa H_2. The thermodynamic functions for Nd–H system are developed based on the critical assessment of the equilibrium pressure P_{H2} over sample at 600–800 °C, invariant reaction temperatures and hydrogen solubilities in Nd. Two sets of thermodynamic functions for Nd–Ni–H system are provided because two sets of models for $NdNi_5$, $NdNi_5H_3$ and $NdNi_5H_6$ are adopted. The $NdNi_5$, $NdNi_5H_3$ and $NdNi_5H_6$ are modeled as separated phases in parameter set 1 and described as a single phase in parameter set 2. The calculated decomposition temperatures of $NdNi_5H_3$ and $NdNi_5Ni_6$ at different pressures agree well with the experimental data. However, the thermodynamic description of parameter set 1 is recommended through comparing the calculated pressure–temperature–composition (PCT) curves with the experimental results. In addition, the obtained thermodynamic database of Nd–Ni–H system is applied to analyze the hydriding process of Nd–Ni alloys.
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1. Introduction

Binary Nd–Ni intermetallic compounds have been investigated frequently because of possible applications as hydrogen storage materials. Commercial nickel-metal hydride battery electrodes utilize the $CaCu_5$-type $RENi_5$ alloys (RE=La, Ce, Pr, and Nd) [1,2]. The Nd–A_2B_7 alloy showed improved high-rate performance and exhibited the ability to maintain a relatively high voltage at a low state-of-charge [3,4]. Iwatake et al. [5] and Iwase et al. [6] investigated the superlattice structures and hydrogen storage property of Nd_2Ni_7. In addition, the RE in RE–Ni alloys are usually substituted by Mg or Al to modify its thermodynamic stability and high hysteresis [2,7]. The substitution of La with Nd in La_2MgNi_9 lowered the thermodynamic stability and allowed the electrochemical discharge capacity of the $La_{1.5}Nd_{0.5}MgNi_9$ hydrides to reach 400 mA h/g [8,9]. It can be seen that the thermodynamic properties of Nd–Ni alloys and the related hydrides are of great importance and become the base to evaluate the hydrogen storage capacities of alloys, thermodynamic stabilities of the candidate hydrides and the reversibility of hydrogen uptake/release reaction. Furthermore, a full understanding of the Nd–Ni–H system is essential to better describe the absorption/desorption processes of multicomponent alloy from thermodynamic aspect. However, there is lack of a complete thermodynamic evaluation of Nd–Ni–H system. Among the RE–Ni–H systems, the La–Ni–H system have been assessed frequently [10,11]. Because the $RENi_5$ alloys have similar structure characteristics, the structure of $LaNi_5H_x$ [12–14] can provide us many useful information about the occupation sites of hydrogen. The details are discussed in literature review. In the present study, a thermodynamic description of the Nd–Ni–H system is provided by critical assessment of the experimental information. The temperature of reaction $NdNi_5H_3 \leftrightarrow NdNi_5 + 1.5H_2$ in the pressure range of 3.01–11.01 MPa is determined by HP-DSC to verify the calculated result.

2. Literature review

2.1. Binary systems

Nd has a hexagonal crystal structure at room temperature and transforms to a body-centered cubic structure at 862 °C [15]. The solubility limits of the metal and hydride phases were

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determined by Peterson [16] through hot vacuum extraction analysis of equilibrated phases. The existence of a metal-rich intermediate phase γ was verified by the discontinuity in the solid-solubility limits at 680 °C [16], but its crystal structure was unclear. Renaudin et al. [17] reported a trigonal NdD_3 (space group *P*–3*c*1), where one site of D is near the center of the tetrahedral interstices, and two sites of D are off-center octahedral interstices with triangular (D3, site 2*a*), or nearly triangular metal co-ordinations (D2, site 4*d*) respectively. The structure of cubic $NdH_{2.61}$ is similar to that of tetragonal $NdH_{2.36}$ [18] except that the hydrogen content of the former is higher than that of the latter and the hydrogen distribution in $NdH_{2.61}$ is less ordered. In NdH_2, the hydrogen atoms occupy tetrahedral interstices with full occupancy and octahedral interstices with partial occupancies [17,19]. The structure of $NdD_{2.27}$ is found to be centrosymmetric where deuterium atoms fill tetrahedral interstices with nearly full occupancy (96%) and one subset of octahedral interstices with partial occupancy (72%) [20]. Most of investigations [17,21–28] indicated the NdH_2 was a fcc phase with space group of *Fm–3m*, but few studies showed that it had a tetragonal structure [17,18,20]. Mintz et al. [29] determined the enthalpy change of the phase transformation $NdH_{2.7}$(hcp)→$NdH_{2.7}$(fcc) at 275 °C on heating was 6.7 kJ/mol. Therefore, the stable phase fcc-NdH_2 was considered in present work. The phases in Nd–H system reported in literature are summarized in Table 1.

The phase transformation from Dhcp(Nd) to Bcc(Nd) was lowered from 860 to 814 °C, and the lower temperature boundary of the bcc phase region increased with increasing hydrogen content to a peritectoid at 880 °C [16]. The phase transformation of metal phase and neodymium hydride to γ phase was found at 680 °C by Peterson [16] and 678 °C by Fujita [39]. Bischof et al. [40] predicted that a peritectic reaction existed in each of the light lanthanide–hydrogen systems based on their measurements in the La–H, Ce–H and Pr–H systems.

The PCT relations of the Nd–H system were studied by Mulford and Holley [19] in the temperature range of 150–800 °C. The enthalpy of formation for NdH_2 is -187.6 ± 3.3 kJ/mol [19] calculated by Van't Hoff equation. But the enthalpy of formation and entropy change for the phase transformation of $Nd + H_2 \leftrightarrow NdH_2$ are -207.2 kJ/mol and -141.9 J/mol K determined by Toguchi et al. [41].

The phase diagram and thermodynamic properties of Ni–H system at pressures up to 50.0 MPa had been well described by the optimized thermodynamic functions provided by Zeng et al. [42]. Because a different lattice stability of liquid hydrogen was adopted by Palumbo et al. [10] in assessment of La–Ni–H system, the binary Ni–H system was partially reassessed. The thermodynamic parameters for Ni–H system [10] were used in our previous thermodynamic reassessment of La–Ni–H system [11]. Therefore, the thermodynamic description for Ni–H system [10] was also adopted in present work.

With regard to the Nd–Ni system, Huang et al. [43] re-modeled this system due to the significant difference of the invariant reaction and congruent melting temperatures between previous data [44] and their results determined from DSC and DTA. However, the calculated enthalpy of formation for $NdNi_5$ at 25 °C was larger than -26.2 ± 1.1 kJ/mol which is determined by Guo and Kleppa [45]. Therefore, the parameter of $NdNi_5$ is modified according to this value.

2.2. Nd–Ni–H system

The hydrogen storage property of $NdNi_5$ has been frequently investigated by researchers [32,46]. Takaguchi and Tanaka [31] determined the hydriding/dehydriding characteristics of $NdNi_5$ and the effect of Sn-substitution. The PCT curves at −20 °C showed two plateaus, which indicated that there are two hydrides $NdNi_5H_{3.6–4.2}$ and $NdNi_5H_{5.4–6.0}$. The enthalpy of formation and entropy change for the $NdNi_5H_{3.6–4.2}$ calculated by Van't Hoff equation were −27.9 kJ/mol and −117.8 J/mol K, respectively. Senoh et al. [1,47] studied the multi pressure plateaus of $RENi_5$ (RE=La, Ce, Pr, Nd, Sm, Gd, Tb, and Dy) alloys in the temperature range of −77 to 150 °C. The hydrides β, γ and δ are corresponding to the solid solution of hydrogen atoms at 3*f*, 6*m* and 4*h* sites in the *P*6/*mmm* space group [1,13].

The crystal structures of $NdNi_5H_3$ and $NdNi_5H_6$ have been seldom discussed. However, because the $RENi_5$ alloys have similar structure characteristics, the structure of $LaNi_5H_x$ [12–14] provides us many useful information. All of $RENi_5$ alloys have a *hP*6–$CaCu_5$

Table 1
The crystal structure and lattice parameter of the phases in Nd–Ni–H system.

Phases	Pearson symbol	Space group	Lattice parameters (Å)			Ref.
			a	*b*	*c*	
Dhcp(Nd)	*hP*4	$P6_3/mmc$ (No.194)	3.6582		11.7966	[22]
Bcc(Nd)	*cI*2	*Im–3m* (No.229)	4.13			[15]
Fcc(Ni)	*cF*4	*Fm–3m* (No.225)	3.523			
Nd_2Ni_{17}	*hP*38	$P6_3/mmc$ (No.194)	8.402		8.408	[30]
$NdNi_5$	*hP*6	*P*6/*mmm* (No.191)	4.949∼4.973		3.974–4.0124	[30–33]
αNd_2Ni_7	*hP*38	$P6_3/mmc$ (No.194)	5.001(1)		24.437(4)	[6]
βNd_2Ni_7	*hP*108	*P*	4.986		72.630	[5]
$NdNi_3$	*hR*12	*R–3m*	4.9918		24.334	[30,34]
$NdNi_2$	*cF*24	*Fd–3m* (No.227)	7.270∼7.265			[35,36]
NdNi	*oC*8	*Cmcm* (No.63)	3.8059	10.462	4.3345	[33]
Nd_7Ni_3	*hP*20	$P6_3mc$ (No.186)	9.8732		6.2977	[37]
Nd_3Ni	*oP*16	*Pnma* (No.62)				[38]
γ						[16]
NdH_2	*cF*4	*Fm–3m* (No.225)	5.431∼5.469			[17,21–28]
	*tI*8	$I4_1md$ (No.109)	5.422		10.908	[18]
	*tI*8	$I4_1/amd$ (No.141)	5.42545(9)		10.8819(4)	[20]
	*hP*6	*P–3c*1 (No.165)	6.6629(3)		6.8853(4)	[17]
$NdNi_5H_3$			4.952		3.988	[31]
$NdNi_5H_6$						

type structure with *P*6/*mmm* space group [33]. Richter et al. [13] calculated the hydrogen jump geometry in $LaNi_5H_x$ and their results showed that the occupancy of 6*m* sites (9%) at 150 °C decreases to 6% at room temperature. The hydrogen atoms mainly occupy 3*f* sites at lower hydrogen concentration and additionally occupy 6*m* sites at higher hydrogen concentration [14]. The capacity of 3*f* sites is corresponding to $LaNi_5H_3$ and the additional capacity of the 6*m* sites is corresponding to $LaNi_5H_9$. However, the spacing of 6*m* sites is about 1.6 Å [48], which is shorter than the minimum distance of 2.1 Å [49] of two hydrogen atoms in a lattice. Therefore, the 6*m* sites are not fully occupied by hydrogen at moderate pressures due to blocking effect of the very closely spaced sites [14]. The occupation of 3*f* sites and partial occupation of 6*m* sites lead to the formation of $LaNi_5H_6$. The stabilities of $RENi_5H_3$ and $RENi_5H_6$ are related to their enthalpy of formation [1]. The $LaNi_5H_3$ would appear at temperature higher than 150 °C reported by Senoh et al. [1] and 80 °C reported by Akiba et al. [12], while the $NdNi_5H_3$ and $NdNi_5H_6$ can be observed in the investigated temperature range of −77 to 50 °C [1].

The crystal structure and hydrogen storage property of Nd_2Ni_7 alloy were investigated by Iwase et al. [6] and the results suggested that the hydrogenation of Nd_2Ni_7 is irreversible. Aoki et al. [50] investigated the hydrogen induced amorphization in $RENi_2$ (RE = Y, La, Ce, Pr, Nd, Sm, Gd, Tb, Dy, Ho, and Er) Laves phases. The result showed that no metal hydride with the $RENi_2$ was observed and the $NdNi_2$ decomposed into NdH_2 and $NdNi_5$ at room temperature and 500 °C. Therefore, it can be deduced that there are only two ternary hydrides, $NdNi_5H_3$ and $NdNi_5H_6$, existing in the Nd–Ni–H system.

3. Experimental procedure

In order to determine the phase transition temperature of reactions $NdNi_5H_3 \leftrightarrow NdNi_5 + 1.5H_2$ and $NdNi_5H_6 \leftrightarrow NdNi_5H_3 + 1.5H_2$, the single phase of $NdNi_5$ was prepared by a medium frequency induction furnace using Nd (≥99.99 wt%) and Ni (≥99.99 wt%) blocks. The sample was re-melted for three times to ensure its homogeneity. Then the ingot was annealed at 800 °C for 19 h to obtain single $NdNi_5$ phase. The phase composition of annealed sample was characterized by X-ray diffraction (XRD) using 18 KW D/MAX2500V+/PC diffractometer with Cu Kα radiation. NETSCH204 HP-DSC was used to examine the phase transition temperatures. The sample firstly absorbed hydrogen under 10.01 MPa H_2 at room temperature for 20 h, then was heated to 130 °C with heating rate of 3 °C/min to release hydrogen, and finally was cooled to room temperature. This process was repeated three times until the DSC curves almost coincided. It indicated that the sample was activated completely. The temperature of phase transition was determined under 3.01, 4.02, 6.01, 8.01, 10.01 and 11.01 MPa H_2 respectively with the heating/cooling rate of 3 °C/min and the flowing rate of 15 mL/min H_2. Before each examination, the sample absorbed hydrogen under each pressure for 20 h to ensure its complete hydrogenation.

4. Experimental results

Fig. 1 shows the XRD pattern of $NdNi_5$ powders. Besides the $NdNi_5$, no any other impurity phase is observed from the XRD pattern. Rietveld refinement is performed using GSAS softwave, and the refined lattice parameters are $a = b = 4.9556$ Å and $c = 3.9810$ Å. The diffraction profile fitted by Rietveld refinement using these parameters is shown in Fig. 1, with the agreement factors of $R_{wp} = 15.36\%$ and $Gof = 1.82$.

The HP-DSC curves under different hydrogen pressures are

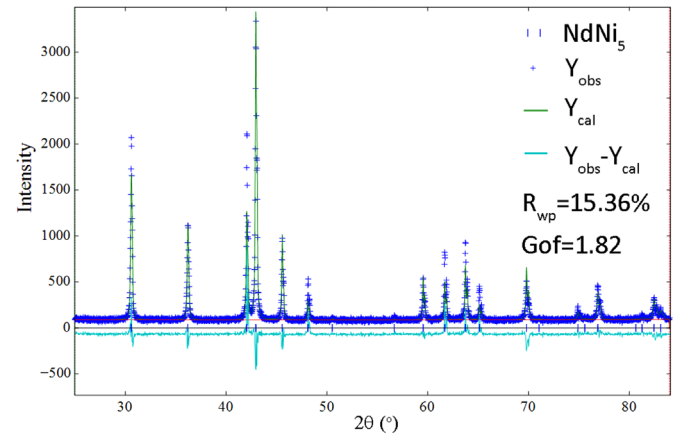

Fig. 1. The Rietveld refinement on the observed XRD pattern of $NdNi_5$: Y_{obs} is the observed diffraction data, Y_{cal} is calculated the diffraction profile, R_{wp} is the weight profile agreement factor, and *Gof* is the goodness of fit indicator.

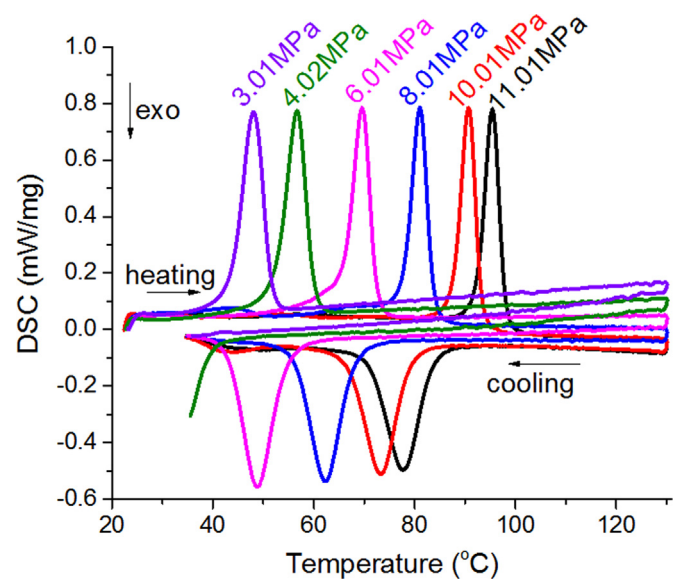

Fig. 2. The HP-DSC curves of $NdNi_5$ powders under 3.01–11.01 MPa H_2.

plotted in Fig. 2. The endothermic and exothermic peaks shift toward high temperature with the increase of hydrogen pressure. Below 4.02 MPa H_2, the exothermic peak cannot be detected because the start temperature is lower than that which can be detected by DSC. The endothermic peaks indicate the release of hydrogen, on the contrary, the exothermic peaks are corresponding to the hydriding of $NdNi_5$. It should be noticed that there is only one peak in whole heating or cooling process, implying that only one phase transition has happened. According to the PCT data [1,31,32], the decomposition pressures of $NdNi_5H_3$ and $NdNi_5H_6$ were 3.0 and 6.8 MPa respectively at about 25 °C. However, only one endothermic/exothermic (dehydriding/hydriding) peak is observed even at 11.01 MPa. Because the activated powder sample was kept at room temperature (22–24 °C) for 20 h, the hydriding time is enough for $NdNi_5$ to transform to its stable hydride. Therefore, it can be conclude that the formation pressure of $NdNi_5H_6$ at 25 °C is larger than 11.01 MPa or the formation temperature of $NdNi_5H_6$ under 11.01 MPa is lower than 25 °C. The determined temperatures of phase transition during heating and cooling processes are listed in Table 2. The determined temperature at 4.01 MPa is close to that

Table 2
The determined phase transition temperature of the reaction $NdNi_5H_3 \leftrightarrow NdNi_5 + 1.5H_2$ compared with calculated results.

Pressure (MPa)		3.01	4.02	6.01	8.01	10.01	11.01
Exp. T (°C)	Heating	41.5	51.4	63.6	75.5	85.1	90.3
	Cooling	–	40.9	54.7	68.0	79.0	83.3
Cal. T (°C)	Parameter set 1	38.4	47.3	60.8	71.1	79.6	83.4
	Parameter set 2	38.3	47.1	60.3	70.3	78.6	82.2

of reaction $NdNi_5H_3 \leftrightarrow NdNi_5 + 1.5H_2$ in literature [1].

The hysteresis behavior is observed, that the dissociation temperature in heating process is higher than the formation temperature in cooling process. The width of hysteresis is related with heating rate and the kinetics of reaction. If the heating rate is fast or the reaction kinetics is slow, the measured hysteresis will be large. In present work, the difference of dissociation temperature and formation temperature is in the range of 6.1–10.5 °C and decreases with the increase of hydrogen pressure. Wicke and Blaurock [51] concluded that the equilibrium plateau pressure coincides with the decomposition plateau pressure for Pd–H system. Zeng et al. [52] found that the experimental dissociation pressures of MgH_2 by various workers agreed very well, whereas the formation pressures are varied. Flanagan and Clewley [53] argued that the true equilibrium plateau pressure lies somewhere in between the plateau formation and decomposition pressures for the same system. Therefore, the equilibrium reaction temperature should lie between the decomposition and formation temperatures, and may be relatively close to decomposition value.

5. Thermodynamic modeling and assessment

The Gibbs free energy of the pure elements is taken from the SGTE compilation by Dinsdale [54]. The Gibbs energy functions for gaseous state Ni and H are obtained from JANAF tables [55]. The Gibbs energy function for gaseous state Nd is obtained from the vapor pressure of the pure element [56] and the Gibbs energy of condensed phases.

The gas phase containing H_2, H, Nd and Ni is considered as an ideal mixture. The molar Gibbs energy of the gas phase is

$$G^{Gas} = \sum_i x_i \left[{}^{\circ}G_i^{Gas} + RT \ln(P/P_0) \right] + RT \sum_i x_i \ln x_i \tag{1}$$

The Liquid (L) phase is modeled as substitutional-solution phases. The Gibbs energy is described by

$$\begin{aligned} G^L = {} & x_{Nd} G_{Nd}^{\circ,Liquid} + x_{Ni} G_{Ni}^{\circ,Liquid} + x_H G_H^{\circ,Liquid} \\ & + RT\left(x_{Nd} \ln x_{Nd} + x_{Ni} \ln x_{Ni} + x_H \ln x_H\right) \\ & + x_{Nd} x_H \sum_{k=0}^{m} {}^{k}L_{Nd,H}^{Liquid} \left(x_{Nd} - x_H\right)^k + x_{Nd} x_{Ni} \sum_{k=0}^{m} {}^{k}L_{Nd,Ni}^{Liquid} \left(x_{Nd} - x_{Ni}\right)^k \\ & + x_{Mg} x_H \sum_{k=0}^{m} {}^{k}L_{Ni,H}^{Liquid} \left(x_{Mg} - x_H\right)^k \end{aligned} \tag{2}$$

where ${}^{k}L_{Nd,H}^{Liquid}$, ${}^{k}L_{Nd,Ni}^{Liquid}$ and ${}^{k}L_{Ni,H}^{Liquid}$ are the binary interaction parameters.

The Dhcp, Bcc and Fcc phases are considered to dissolve hydrogen interstitially. Those phases are described using two sublattices $(Nd, Ni)_1(H, Va)_n$, where the first sublattice represents the partial substitutional-solution of Nd and Ni, and the second sublattice represents the interstitial sites occupied by hydrogen atoms and vacancies. The stoichiometry n is related with the number of occupied interstitial sites per Nd and Ni atoms. It seems that the evidence favors tetrahedral site occupation for hydrogen in Bcc and Fcc forms but the situation regarding Hcp form is less clear [57,58]. The occupation of tetrahedral site is assumed for Dhcp. Assuming tetrahedral site for Bcc, the value of n should be 6. Königsberger et al. [59] disscussed the large discrepancy between geometric value and the actual n used for Bcc form. The reason of large value of n for Bcc is not chosen might because of the site blocking effect and thermodynamic self-consistency as extrapolating to multicomponent system. In present work, the n is assumed to 3 for Bcc. The structure of γ was not reported. The two-sublattice model $Nd_1(H, Va)_2$ as same as that for Dhcp phase is used. According to the structure information in the literature review, it can be concluded that the stable hydride is fcc structure and use NdH_2 to label it. The NdH_2 is modeled as three-sublattice model $(Nd)_1(H, Va)_2(H, Va)_1$ due to the two kinds of interstices which can be occupied by H atoms [17,19]. The $(H,Va)_2$ represents the tetrahedral interstices and $(H,Va)_1$ represents the octahedral interstices.

Since few experimental evidence of the solubility of hydrogen in Nd–Ni binary solid phases exists except $NdNi_5$, those compounds are considered as stoichiometric phases. Due to the structure characteristics of $NdNi_5$, $NdNi_5H_3$ and $NdNi_5H_6$, two set of models are employed to describe the three phases. In parameter set 1, the three phases are modeled by using three separated models. $NdNi_5$ is modeled as $(Nd)_1(Ni)_5(H, Va)_3$, where $(H, Va)_3$ represents hydrogen partially occupying the 3f interstices. Both $NdNi_5H_3$ and $NdNi_5H_6$ are modeled using $(Nd)_1(Ni)_5(H)_3(H, Va)_6$ formula, where $(H)_3$ represents hydrogen fully occupying the 3f interstices and $(H, Va)_6$ represents hydrogen partially occupying the 6m interstices. The same thermodynamic model are used for $NdNi_5H_3$ and $NdNi_5H_6$ because their crystal structure are similar. If the hydrogen atoms occupied the same interstices with same fraction, the Gibbs free energy of formation should be same. However, the interaction parameters describing the two phases are different to reveal the stable composition range of the two phases. In parameter set 2, the three phases are described as a single phase, $(Nd)_1(Ni)_5(H, Va)_3(H, Va)_6$, consistently with their crystal structures. The Gibbs free energy of those phases are described using compound energy formalism. Taking the $NdNi_5H_6$ in parameter set 1 as an example, the molar Gibbs energy can be expressed as

$$\begin{aligned} G^{NdNi_5H_6} = {} & y'_{Nd} y''_{Ni} y'''_H y''''_H G_{Nd:Ni:H:H}^{NdNi_5H_6} \\ & + y'_{Nd} y''_{Ni} y'''_H y''''_{Va} G_{Nd:Ni:H:Va}^{NdNi_5H_6} \\ & + \frac{6}{15} RT \left(y'''_H \ln y'''_H + y''''_{Va} \ln y''''_{Va} \right) \\ & + y'_{Nd} y''_{Ni} y'''_H y''''_H y''''_{Va} \sum_{k=0}^{m} {}^{k}L_{Nd:Ni:H:H,Va}^{NdNi_5H_6} \left(y'''_H - y''''_{Va} \right) \end{aligned} \tag{3}$$

where y'_{Nd} is the site fraction of Nd in the first sublattice, y''_{Ni} is the site fraction of Ni in the second sublattice, y'''_H is the site fraction of H in the third sublattice, y''''_H and y''''_{Va} are the site fractions of H and Va in the fourth sublattice.

The lattice stabilities of NdH_2 are assessed based on the enthalpy of formation [19,41] and the equilibrium pressure P_{H2}. In order to avoid its excessive stability at room temperature, the enthalpy of formation of -187.6 ± 3.3 kJ/mol [19] is accepted. The parameters for Dhcp, γ, Bcc are determined by optimizing the phase boundaries of γ/NdH_2, Dhcp/NdH_2, γ/Bcc and Bcc/Dhcp [16,19] and the temperatures of invariant reactions [16]. The calculated enthalpy of formation for $NdNi_5$ at 25 °C in Huang et al.' assessment [43] is larger than -26.2 ± 1.1 kJ/mol reported by Guo and Kleppa [45], which makes the equilibrium pressure P_{H2} over $NdNi_5$ (dilute phase) relatively low with respect to experimental data [1,31] when the hydrogen solubility is matched. Therefore, the partial reassessment of the Nd–Ni system, limited to $NdNi_5$,

Nd_2Ni_{17}, Nd_2Ni_7, $NdNi_3$ and $NdNi_2$, is carried out. The lattice stabilities of $NdNi_5H_3$ and $NdNi_5H_6$ are assessed based on the enthalpy of formation and the interaction parameters are determined based on the equilibrium pressure P_{H2} over sample. Because of the hystereis phenomenon, the dissociation temperatures or pressures are considered as optimizing the parameters of phases. Furthermore, a little deviation in the hysteresis range is also allowed. The PANDAT software [60] is used for thermodynamic optimization. The optimized parameters are listed in Appendix.

6. Calculated results and discussion

The calculated phase diagram of Nd–H system is shown in Fig. 3(a). The temperatures of reactions and composition of phases are listed in Table 3 compared with experimental data. It suggests that the invariant reaction types and temperatures are in a good agreement with the experimental data. The calculated phase boundaries of Bcc/Dhcp, γ/Bcc, γ/NdH_2 are in satisfactory agreement with the limited experimental data [16,19]. The calculated solubilities of H in Dhcp deviate a little from experimental data toward high hydrogen concentration, but the calculated equilibrium pressure P_{H2} at 600–800 °C is in accordance with the data reported by Mulford [19] (shown in Fig. 3(b)). The equilibrium pressure P_{H2} data is adopted because the boundary of dilute phase/dilute phase + hydride is difficult to identify, while the experimental accuracy of equilibrium pressure is high. The calculated enthalpies of formation of NdH_2 are −187.4 kJ/mol H_2 at 25 °C, −182.3 kJ/mol H_2 at 200 °C and −169.1 kJ/mol H_2 at 600 °C, respectively. They are close to the values obtained from PCT data [19,41].

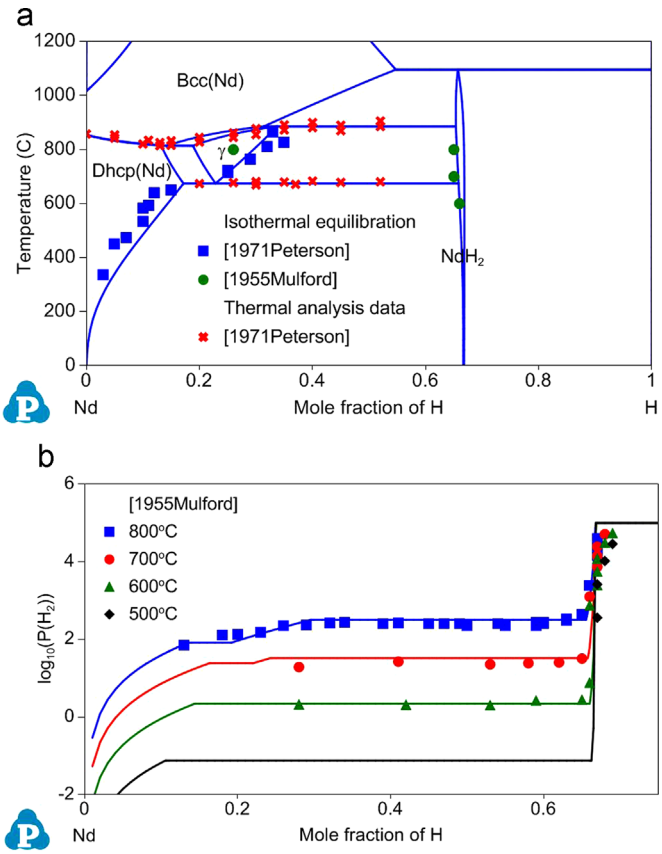

Fig. 3. (a) The calculated Nd–H phase diagram under 0.1 MPa, and (b) the calculated PCT curves in Nd–H system.

Table 3
The invariant reactions in the Nd–H system.

Invariant reactions	Temperature (°C)		Phase name	x_H in phases (at)
	Literature	Calculated		
$NdH_{2.61}$ + Bcc (Nd) → γ	880 [16, 61], 881 [39], 885 [40]	885.6	$NdH_{2.61}$ Bcc(Nd) γ	0.6535 0.3135 0.3371
Bcc(Nd) → γ + Dhcp(Nd)	814 [16, 61], 813 [39], 819 [40]	814.4	Bcc(Nd) γ Dhcp(Nd)	0.1423 0.1885 0.1322
γ → $NdH_{2.61}$ + Dhcp(Nd)	680 [16, 61], 678 [39], 670 [40]	674.6	γ $NdH_{2.61}$ Dhcp(Nd)	0.2285 0.6581 0.1718

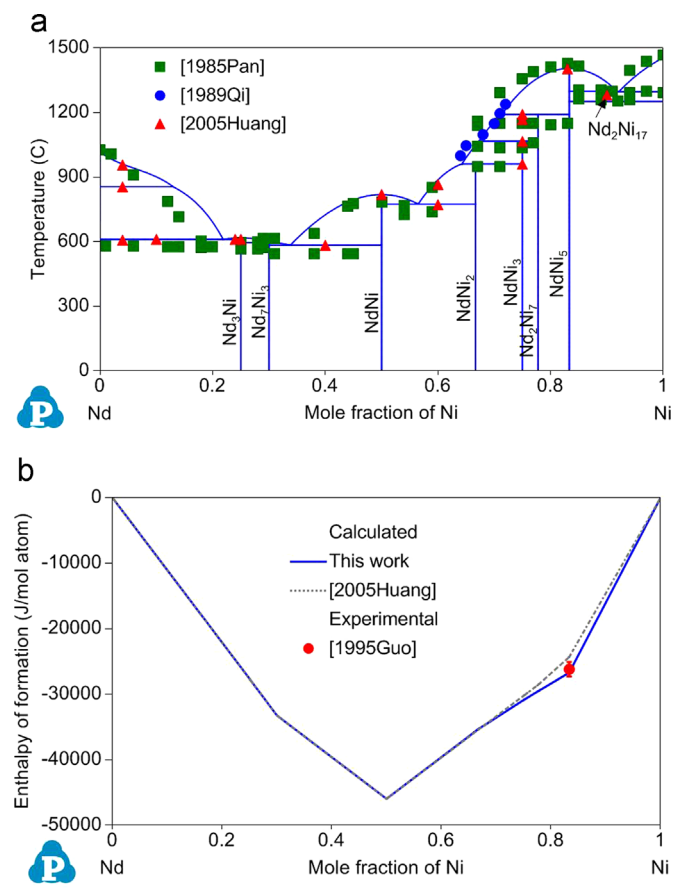

Fig. 4. (a) The calculated phase diagram of Nd–Ni system compared with experimental data, and (b) the calculated enthalpies of formation of Nd–Ni alloys at 25 °C.

The Nd–Ni system is partially assessed because the calculated enthalpy of formation of $NdNi_5$ [43] a little deviate from the experimental value [45]. The calculated phase diagram and enthalpies of formation of the Nd–Ni alloys at 25 °C is shown in Fig. 4. The calculated temperatures of invariant reactions and phase boundaries of liquid/solid phases are in a good agreement with experimental data. The calculated enthalpy of formation of $NdNi_5$ is −26.7 kJ/mol, which agrees with −26.2 ± 1.1 kJ/mol [45] in literature.

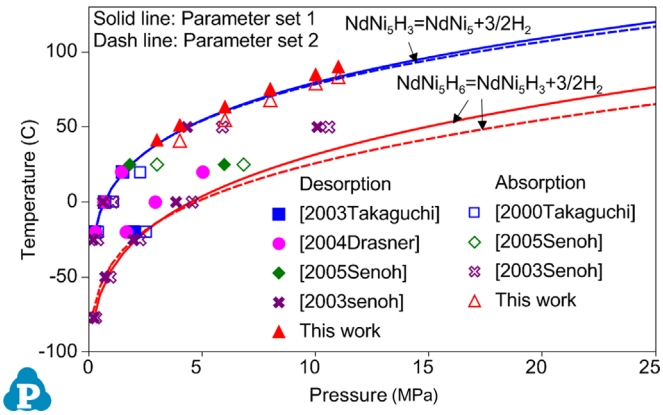

Fig. 5. The calculated equilibrium decomposition temperatures and pressures of $NdNi_5H_3$ and $NdNi_5H_6$ from the two sets of parameters compared with experimental data.

The relationships between hydrogen pressure and decomposition temperature of $NdNi_5H_3$ and $NdNi_5H_6$ were considered when optimizing the interaction parameters of the phases. The calculated temperatures of decompositions versus hydrogen pressure are plotted in Fig. 5 using the two thermodynamic databases. Both calculated curves from parameter set 1 and 2 for $NdNi_5H_3 \leftrightarrow NdNi_5 + 1.5H_2$ reaction agree well with the experimental data determined in present work as well as the data derived from PCT [1,31,32,47]. However, the temperature of the $NdNi_5H_6 \leftrightarrow NdNi_5H_3 + 1.5H_2$ reaction under 5–11 MPa is lower than the experimental data [1,32,47]. It is because the temperature for the reaction determined in present work is lower than 40.9 °C in the pressure range of 3.01–10.01 MPa. The decomposition temperature below 5 MPa is consistent with the experimental data. The calculated temperatures of the $NdNi_5H_3 \leftrightarrow NdNi_5 + 1.5H_2$ reaction are listed in Table 2 compared with the determined results. The calculated reaction temperature from parameter set 2 is close to that from parameter set 1 and agree with the experimental data. But the difference between the calculated values from the two databases becomes obviously under higher pressure. Comparing the calculated reaction temperature with the experimental data in Table 2, the calculated value from parameter set 1 shows better agreement than that from parameter set 2.

The solubilities of hydrogen in $NdNi_5$, $NdNi_5H_3$ and $NdNi_5H_6$ are taken from the PCT data [1,31,32,47] and considered in the assessment of the $NdNi_5$–H isolethal section. The calculated PCT curves at different temperatures are shown in Fig. 6 compared with the experimental data. The calculated plateau pressures and equilibrium pressures over $NdNi_5H_3$ and $NdNi_5H_6$ agree well with the experimental points. However, the calculated pressures over $NdNi_5$ are lower than the experimental values at 0–50 °C. This deviation comes from the relatively high Gibbs free energy of $NdNi_5$, although the enthalpy of formation of $NdNi_5$ has been lowered to −26.7 kJ/mol. In order to fit the plateau pressures of $NdNi_5 + NdNi_5H_3$, a large value ${}^1L^{NdNi_5}_{Nd:Ni:H,Va}$ should be used. Therefore, the calculated hydrogen chemical potential near $NdNi_5$ is relatively low. With regard to the equilibrium pressure over $NdNi_5H_6$ single phase, the calculated pressure from parameter set 1 is closer to experimental points than that from parameter set 2. Therefore, the parameter set 1 is recommended to describe the Nd–Ni–H system. The calculated enthalpies of formation of $NdNi_5H_3$ and $NdNi_5H_6$ from parameter set 1 are listed in Table 4 compared with the measured ones, which suggests that a good agreement is obtained.

The isothermal section of Nd–Ni–H system at 25 °C under 10 MPa and the isopleth of the $NdNi_5$–H section are calculated using the parameter set 1 and shown in Fig. 7. The red dash lines shown in Fig. 7(b) is the contour lines of pressure. It suggests the equilibrium pressures over different phases. The isopleth of $NdNi_5$–H calculated from parameter set 2 is similar with Fig. 7(b) and not shown in present work. The miscibility gap does not appear.

With guide of the thermodynamic database, the hydrogen storage capacity and PCT curves for any other Nd–Ni alloys can be easily predicted through calculation. Taken the $NdNi_2$ as an example, the predicted PCT curve at 25 °C is shown in Fig. 8. The

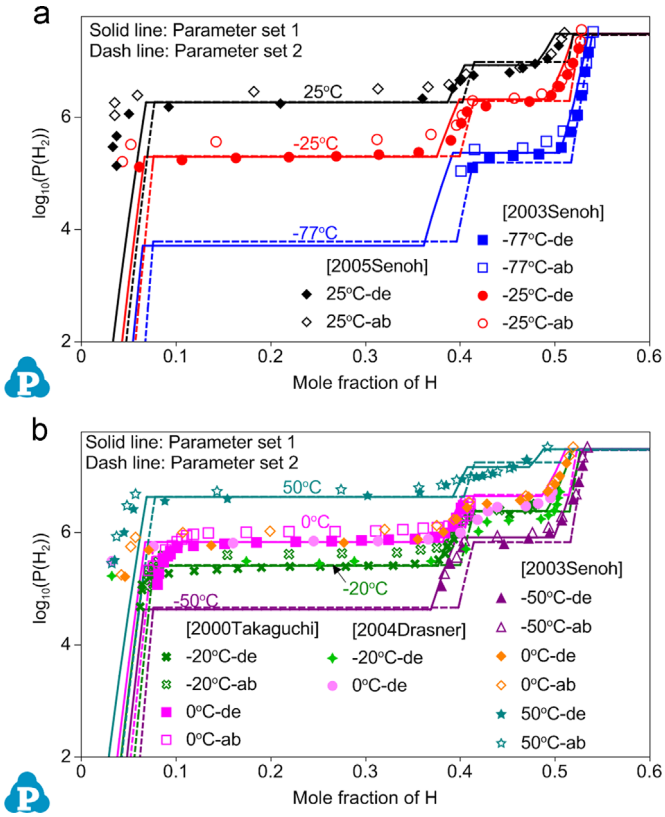

Fig. 6. The calculated PCT curves using the two sets of parameters at different temperatures compared with experimental data.

Table 4
The enthalpy of formation of the hydrides calculated from parameter set 1 compared with measured values.

Phases	ΔH_f (kJ/mol H_2)		Notes	Ref.
	Calculated	Experimental		
NdH_2	−169.1, 600 °C	−187.6	PCT, 600–800 °C	[19]
	−182.3, 200 °C	−207.2	PCT, 200–950 °C	[41]
$NdNi_5H_3$	−105.4, 25 °C	−106.5[a]	PCT, −20–20 °C	[31]
		−106.3[a]	PCT, −25–50 °C	[1]
$NdNi_5H_6$	−80.5, 25 °C	−76.4[a]	PCT, −20–20 °C	[31]
		−77.1[a]	PCT, −20–20 °C	[32]
		−68.3[a]	PCT, −77–50 °C	[1]

[a] Original data are for the hydrogenation reaction from the binary $NdNi_5$. They have been converted into present values using the enthalpy of formation of $NdNi_5$ (−26.2 ± 1.1 kJ/mol [45]).

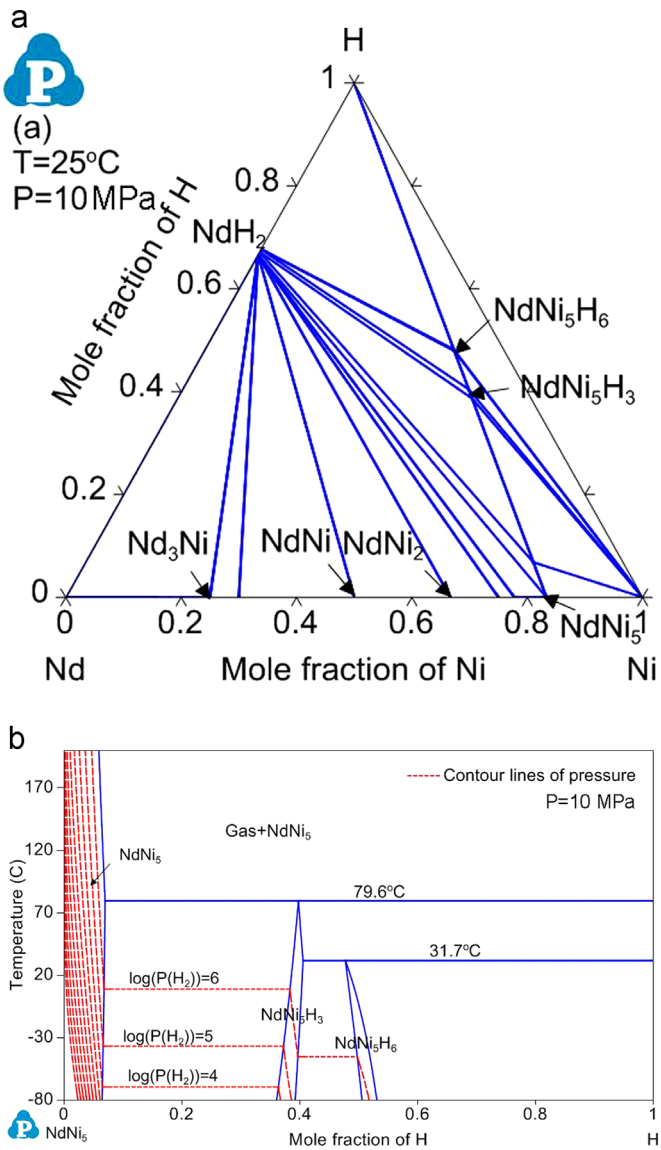

Fig. 7. (a) The isothermal section of Nd–Ni–H system at 25 °C under 10 MPa, and (b) the isopleth of $NdNi_5$–H section under 10 MPa calculated by parameter set 1. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

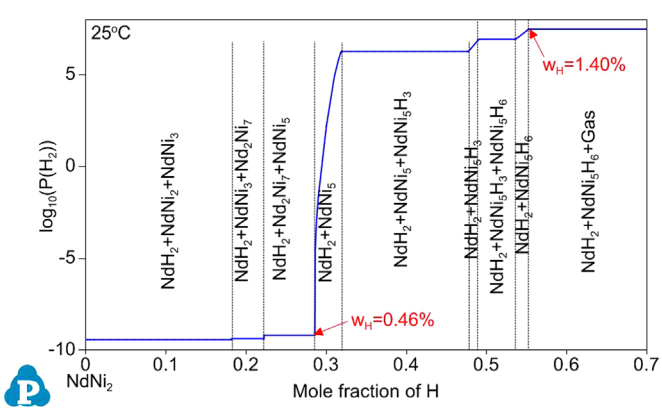

Fig. 8. The predicted PCT curve of $NdNi_2$ at 25 °C using parameter set 1.

hydrogenation process of $NdNi_2$ can be separated into 8 steps: (a) Nd in $NdNi_2$ reacting with hydrogen leads to $NdNi_2$ decomposition. (b) The $NdNi_3$ further decomposes to NdH_2 and Nd_2Ni_7. (c) The Nd_2Ni_7 further decomposes to NdH_2 and $NdNi_5$. (d) The hydrogen atoms dissolve in $NdNi_5$. (e) $NdNi_5$ absorbs hydrogen to become $NdNi_5H_3$. (f) The hydrogen atoms dissolve in $NdNi_5H_3$. (g) $NdNi_5H_3$ absorbs hydrogen to become $NdNi_5H_6$. (h) Hydrogen atoms dissolve in $NdNi_5H_6$. All of the steps shown in Fig. 8 can be summarized as

$$\begin{aligned}
&NdNi_2 \underset{(a)}{\overset{+H_2}{\rightarrow}} \frac{1}{3}NdH_2 + \frac{2}{3}NdNi_3 \underset{(b)}{\overset{+H_2}{\rightarrow}} \frac{3}{7}NdH_2 + \frac{2}{7}Nd_2Ni_7 \\
&\qquad \underset{(c)}{\overset{+H_2}{\rightarrow}} \frac{3}{5}NdH_2 + \frac{2}{5}NdNi_5 \\
&\underset{(d)}{\overset{+H_2}{\longleftrightarrow}} \frac{3}{5}NdH_2 + \frac{2}{5}NdNi_5(H_x) \underset{(e)}{\overset{+H_2}{\longleftrightarrow}} \frac{3}{5}NdH_2 + \frac{2}{5}NdNi_5H_3 \underset{(f)}{\overset{+H_2}{\longleftrightarrow}} \\
&\qquad \frac{3}{5}NdH_2 + \frac{2}{5}NdNi_5H_{3+x} \\
&\underset{(g)}{\overset{+H_2}{\longleftrightarrow}} \frac{3}{5}NdH_2 + \frac{2}{5}NdNi_5H_6 \underset{(h)}{\overset{+H_2}{\longleftrightarrow}} \frac{3}{5}NdH_2 + \frac{2}{5}NdNi_5H_{6+x}
\end{aligned} \tag{4}$$

The maximum hydrogen storage capacity for the total process is 1.40 wt%. However, the NdH_2 is hard to release hydrogen except that hydrogen pressure is very low ($\log_{10}(P_{H2}) < -4.45$ according to this calculation). Therefore, the steps (a)–(c) occur on the first hydrogenation, and the steps (d)–(h) are reversible in the next hydriding and dehydriding process. The hydrogen storage content for NdH_2 is 0.46 wt%. Thus the reversible hydrogen storage capacity for $NdNi_2$ is 0.94 wt%.

7. Conclusions

The Nd–H phase diagram is assessed and the calcualted results agree well with the experimental data. Based on the sub-binary systems, the thermodynamic functions for Nd–Ni–H system are developed through critical assessment of the equilibrium pressure P_{H2} at different temperatures and the invariant reaction temperatures. Two sets of parameters are provided to describe the $NdNi_5$, $NdNi_5H_3$ and $NdNi_5H_6$. The calculated decomposition temperatures of $NdNi_5H_3$ and $NdNi_5H_6$ by both parameter sets agree with the reaction temperatures determined by HP-DSC. But the PCT curves calcualted by parameter set 1 show better agreement than that by parameter set 2 as comparing with the PCT data in literature. The obtained thermodynamic database of Nd–Ni–H system can be used to predict the hydrogenation behavior and hydrogen storage capacities of other Nd–Ni alloys.

Acknowledgments

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Appendix. The optimized thermodynamic parameters in Nd–Ni–H system

Gas: Model (H_2, H, Nd, Ni, and H)	Note	Ref.
$G_{H_2}^{Gas} = {}^{\circ}G_{H_2}^{Gas} + RT\ln(9.8692327 \times 10^{-6} \cdot P)$		
$G_{H}^{Gas} = {}^{\circ}G_{H}^{Gas} + RT\ln(9.8692327 \times 10^{-6} \cdot P)$		
$G_{Nd}^{Gas} = {}^{\circ}G_{Nd}^{Gas} + RT\ln(9.8692327 \times 10^{-6} \cdot P)$		
$G_{Ni}^{Gas} = {}^{\circ}G_{Ni}^{Gas} + RT\ln(9.8692327 \times 10^{-6} \cdot P)$		
Liquid: Model (Nd, Ni, and H)		
$G_{Nd}^{L} = {}^{\circ}G_{Nd}^{Liquid}$		[43]
$G_{Ni}^{L} = {}^{\circ}G_{Ni}^{Liquid}$		[43]
$G_{H}^{L} = {}^{\circ}G_{H}^{Liquid}$		[10]
${}^{0}L_{Nd,Ni}^{L} = -166540 + 38.798T$		[43]
${}^{1}L_{Nd,Ni}^{L} = 34987 - 5.354T$		[43]
${}^{2}L_{Nd,Ni}^{L} = 24991$		[43]
${}^{0}L_{Ni,H}^{L} = -8690.93694 + 10.7429419T$		[10]
Dhcp: Model $(Nd, Ni)_1(H, Va)_2$		
$G_{Nd:Va}^{Dhcp} = {}^{\circ}G_{Nd}^{Dhcp}$		[43]
$G_{Ni:Va}^{Dhcp} = {}^{\circ}G_{Ni}^{Dhcp} + 500$		[43]
$G_{Nd:H}^{Dhcp} = {}^{\circ}G_{Nd}^{Dhcp} + {}^{\circ}G_{H_2}^{Gas} - 146740.654 + 118.589T$		This work
${}^{0}L_{Nd,Ni:Va}^{Dhcp} = -10000$		[43]
Bcc: Model $(Nd, Ni)_1(H, Va)_3$		
$G_{Nd:Va}^{Bcc} = {}^{\circ}G_{Nd}^{Bcc}$		[43]
$G_{Ni:Va}^{Bcc} = {}^{\circ}G_{Ni}^{Bcc}$		[43]
$G_{Nd:H}^{Bcc} = {}^{\circ}G_{Nd}^{Bcc} + 1.5\,{}^{\circ}G_{H_2}^{Gas} - 135207.418 + 108.37T$		This work
${}^{0}L_{Nd,Ni:Va}^{Bcc} = -8500$		[43]
Fcc: Model $(Nd, Ni)_1(H, Va)_1$		
$G_{Ni:Va}^{Fcc} = {}^{\circ}G_{Ni}^{Fcc}$		[43]
$G_{Nd:Va}^{Fcc} = {}^{\circ}G_{Nd}^{Fcc}$		[43]
$G_{Ni:H}^{Fcc} = {}^{\circ}G_{Ni}^{Fcc} + 0.5\,{}^{\circ}G_{H_2}^{Gas} - 3120.7324 + 180.5062T - 16.255T\ln T + 6.052879 \times 10^{-4}T^2$		[43]
$G_{Nd:H}^{Fcc} = {}^{\circ}G_{Nd}^{Fcc} + 0.5\,{}^{\circ}G_{H_2}^{Gas} + 50000$		This work
${}^{0}L_{Nd,Ni:Va}^{Fcc} = 100000$		[43]
${}^{0}T_{CNi:Va}{}^{Fcc} = 633$		[42]
${}^{0}\beta_{Ni:Va}^{Fcc} = 0.52$		[42]
γ: Model $(Nd)_1(H, Va)_2$		
$G_{Nd:Va}^{\gamma} = {}^{\circ}G_{Nd}^{Dhcp} + 800$		This work
$G_{Nd:H}^{\gamma} = {}^{\circ}G_{Nd}^{Dhcp} + {}^{\circ}G_{H_2}^{Gas} - 138675.944 + 104.16T$		This work
NdH_2: Model $(Nd)_1(H, Va)_2(H, Va)_1$		
$G_{Nd:H:H}^{NdH_2} = {}^{\circ}G_{Nd}^{Dhcp} + 1.5\,{}^{\circ}G_{H_2}^{Gas} - 186000 + 157T$		This work
$G_{Nd:H:Va}^{NdH_2} = {}^{\circ}G_{Nd}^{Dhcp} + {}^{\circ}G_{H_2}^{Gas} - 187451.375 + 124.5T$		This work
$G_{Nd:Va:Va}^{NdH_2} = {}^{\circ}G_{Nd}^{Dhcp} + 50000$		This work
$G_{Nd:Va:H}^{NdH_2} = {}^{\circ}G_{Nd}^{Dhcp} + 0.5\,{}^{\circ}G_{H_2}^{Gas} - 100000 + 140T$		This work

Nd_3Ni: Model $(Nd)_{0.75}(Ni)_{0.25}$

$G^{Nd_3Ni}_{Nd:Ni} = 0.75\ {}^{\circ}G^{Dhcp}_{Nd} + 0.25\ {}^{\circ}G^{Fcc}_{Ni} - 28093 + 7.967T$ [43]

Nd_7Ni_3: Model $(Nd)_{0.7}(Ni)_{0.3}$

$G^{Nd_7Ni_3}_{Nd:Ni} = 0.7\ {}^{\circ}G^{Dhcp}_{Nd} + 0.3\ {}^{\circ}G^{Fcc}_{Ni} - 33789 + 10.595T$ [43]

NdNi: Model $(Nd)_{0.5}(Ni)_{0.5}$

$G^{NdNi}_{Nd:Ni} = 0.5\ {}^{\circ}G^{Dhcp}_{Nd} + 0.5\ {}^{\circ}G^{Fcc}_{Ni} - 46879 + 12.065T$ [43]

$NdNi_2$: Model $(Nd)_{0.3333}(Ni)_{0.6667}$

$G^{NdNi_2}_{Nd:Ni} = 0.3333\ {}^{\circ}G^{Dhcp}_{Nd} + 0.6667\ {}^{\circ}G^{Fcc}_{Ni} - 36591.721 + 4.306T$ This work

$NdNi_3$: Model $(Nd)_{0.25}(Ni)_{0.75}$

$G^{NdNi_3}_{Nd:Ni} = 0.25\ {}^{\circ}G^{Dhcp}_{Nd} + 0.75\ {}^{\circ}G^{Fcc}_{Ni} - 32262.113 + 3.493T$ This work

Nd_2Ni_7: Model $(Nd)_{0.2222}(Ni)_{0.7778}$

$G^{Nd_2Ni_7}_{Nd:Ni} = 0.2222\ {}^{\circ}G^{Dhcp}_{Nd} + 0.7778\ {}^{\circ}G^{Fcc}_{Ni} - 30893.89 + 3.506T$ This work

Nd_2Ni_{17}: Model $(Nd)_{0.1053}(Ni)_{0.8947}$

$G^{Nd_2Ni_{17}}_{Nd:Ni} = 0.1053\ {}^{\circ}G^{Dhcp}_{Nd} + 0.8947\ {}^{\circ}G^{Fcc}_{Ni} - 16387.5 + 1.62603T$ This work

$NdNi_5$: Model $(Nd)_{0.1667}(Ni)_{0.8333}(H, Va)_{0.5}$ Para. Set 1

$G^{NdNi_5}_{Nd:Ni:Va} = 0.1667\ {}^{\circ}G^{Dhcp}_{Nd} + 0.8333\ {}^{\circ}G^{Fcc}_{Ni} - 28248.82 + 4.088T$ Para. Set 1 This work

$G^{NdNi_5}_{Nd:Ni:H} = 0.1667\ {}^{\circ}G^{Dhcp}_{Nd} + 0.8333\ {}^{\circ}G^{Fcc}_{Ni} + 0.25\ {}^{\circ}G^{Gas}_{H_2} - 36328 + 36T$ Para. Set 1 This work

${}^{0}L^{NdNi_5}_{Nd:Ni:H,Va} = 10000$ Para. Set 1 This work

${}^{1}L^{NdNi_5}_{Nd:Ni:H,Va} = 18000$ Para. Set 1 This work

$NdNi_5H_3$: Model $(Nd)_{0.1667}(Ni)_{0.8333}(H)_{0.5}(H, Va)_1$ Para. Set 1

$G^{NdNi_5H_3}_{Nd:Ni:H:Va} = 0.1667\ {}^{\circ}G^{Dhcp}_{Nd} + 0.8333\ {}^{\circ}G^{Fcc}_{Ni} + 0.25\ {}^{\circ}G^{Gas}_{H_2} - 36328 + 36T$ Para. Set 1 This work

$G^{NdNi_5H_3}_{Nd:Ni:H:H} = 0.1667\ {}^{\circ}G^{Dhcp}_{Nd} + 0.8333\ {}^{\circ}G^{Fcc}_{Ni} + 0.75\ {}^{\circ}G^{Gas}_{H_2} - 41168.616 + 87.596T$ Para. Set 1 This work

${}^{0}L^{NdNi_5H_3}_{Nd:Ni:H:H,Va} = -423.529 - 2.941T$ Para. Set 1 This work

${}^{1}L^{NdNi_5H_3}_{Nd:Ni:H:H,Va} = 4000$ Para. Set 1 This work

$NdNi_5H_6$: Model $(Nd)_{0.1667}(Ni)_{0.8333}(H)_{0.5}(H, Va)_1$ Para. Set 1

$G^{NdNi_5H_6}_{Nd:Ni:H:Va} = 0.1667\ {}^{\circ}G^{Dhcp}_{Nd} + 0.8333\ {}^{\circ}G^{Fcc}_{Ni} + 0.25\ {}^{\circ}G^{Gas}_{H_2} - 36328 + 36T$ Para. Set 1 This work

$G^{NdNi_5H_6}_{Nd:Ni:H:H} = 0.1667\ {}^{\circ}G^{Dhcp}_{Nd} + 0.8333\ {}^{\circ}G^{Fcc}_{Ni} + 0.75\ {}^{\circ}G^{Gas}_{H_2} - 41168.616 + 87.596T$ Para. Set 1 This work

${}^{0}L^{NdNi_5H_6}_{Nd:Ni:H:H,Va} = -5107.104 - 14.408T + 0.03488T^{2}$ Para. Set 1 This work

${}^{1}L^{NdNi_5H_6}_{Nd:Ni:H:H,Va} = -15764.706 + 29.41T$ Para. Set 1 This work

$NdNi_5H_x$: Model $(Nd)_{0.1667}(Ni)_{0.8333}(H, Va)_{0.5}(H, Va)_1$ Para. Set 2

$G^{NdNi_5H_x}_{Nd:Ni:Va:Va} = 0.1667\ {}^{\circ}G^{Dhcp}_{Nd} + 0.8333\ {}^{\circ}G^{Fcc}_{Ni} - 28248.82 + 4.088T$ Para. Set 2 This work

$G^{NdNi_5H_x}_{Nd:Ni:H:Va} = 0.1667\ {}^{\circ}G^{Dhcp}_{Nd} + 0.8333\ {}^{\circ}G^{Fcc}_{Ni} + 0.25\ {}^{\circ}G^{Gas}_{H_2} - 36128 + 36T$ Para. Set 2 This work

$G^{NdNi_5H_x}_{Nd:Ni:Va:H} = 0.1667\ {}^{\circ}G^{Dhcp}_{Nd} + 0.8333\ {}^{\circ}G^{Fcc}_{Ni} + 0.5\ {}^{\circ}G^{Gas}_{H_2} - 41820.4 + 59.8T$ Para. Set 2 This work

$G^{NdNi_5H_x}_{Nd:Ni:H:H} = 0.1667\ {}^{\circ}G^{Dhcp}_{Nd} + 0.8333\ {}^{\circ}G^{Fcc}_{Ni} + 0.75\ {}^{\circ}G^{Gas}_{H_2} - 45390.2 + 97.4T$	Para. Set 2	This work
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${}^{0}L^{NdNi_5H_x}_{Nd:Ni:H,Va:H} = -3000$	Para. Set 2	This work
${}^{1}L^{NdNi_5H_x}_{Nd:Ni:H:H,Va} = 11000$	Para. Set 2	This work
${}^{0}L^{NdNi_5H_x}_{Nd:Ni:Va:H,Va} = 23000$	Para. Set 2	This work

Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.calphad.2015.10.009.

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